



MONTE CARLO-BASED STOCHASTIC PORTFOLIO RISK MODELING

Multivariate Covariance Estimation, Tail Risk Quantification, and
Correlation Stress Testing Using Real Market Data



DEVANG NAGAR
2022B4A31042H

Table of Contents

ABSTRACT.....	2
SECTION 1: INTRODUCTION	3
SECTION 2: SYSTEM DESCRIPTION	4
SECTION 3: MATHEMATICAL MODELING OF ASSET RETURNS	5
SECTION 4: MONTE CARLO SIMULATION FRAMEWORK	7
SECTION 5: RISK MEASURES.....	8
SECTION 6: STRESS TESTING AND CORRELATION EFFECTS.....	9
SECTION 7: SIMULATION PROCEDURE.....	10
SECTION 8: RESULTS.....	11
SECTION 9: CONCLUSION.....	13
SECTION 10: PRACTICAL RELEVANCE AND APPLICATIONS	14
SECTION 11: MATLAB IMPLEMENTATION	15
REFERENCES:	17

ABSTRACT

This project focuses on analyzing portfolio risk using Monte Carlo simulation applied to real market data. Daily returns of multiple stocks are treated as a stochastic process, where both individual volatility and correlation between assets are taken into an account. When multiple correlated assets are involved, it becomes difficult to describe portfolio losses using closed-form expressions, which motivates the use of Monte Carlo simulation. The simulation generates a large number of possible one-day market scenarios, from which the distribution of portfolio losses is obtained. From this loss distribution, Value-at-Risk and Expected Shortfall are computed to quantify downside risk at high confidence levels. To study how risk changes under adverse conditions, asset correlations are increased artificially and the impact on tail losses is examined. The results show that higher correlation significantly increases extreme losses and reduces the benefits of diversification. Expected Shortfall is found to be more sensitive to stressed conditions than Value-at-Risk, as it captures the severity of losses beyond the risk threshold. Overall, the project demonstrates how stochastic modeling and simulation can be used to understand and quantify portfolio risk in a practical setting.

SECTION 1: INTRODUCTION

Financial markets are inherently uncertain, and asset prices fluctuate continuously due to a variety of economic and behavioral factors. While historical data provides insight into past performance, it does not directly indicate the range of possible future outcomes, especially during extreme market conditions. Therefore, understanding and quantifying uncertainty is an essential component of portfolio risk analysis.

In many cases, investors focus on average returns or expected performance. However, from a risk management perspective, rare but severe losses are often more important than average outcomes. Even if such extreme events occur infrequently, their financial impact can be significant. This motivates the need to study the entire distribution of portfolio returns rather than relying solely on point estimates.

When constructing a portfolio consisting of multiple assets, overall risk depends not only on the volatility of individual assets but also on how these assets move relative to each other. Correlation between asset returns plays a crucial role in determining whether diversification reduces or amplifies risk. During normal market conditions, diversification can help smooth fluctuations. However, during stressed conditions, assets often move together, leading to amplified losses.

Due to the presence of multiple correlated risk factors, analytical evaluation of portfolio loss becomes increasingly complex, especially when focusing on tail behavior. For this reason, simulation-based approaches provide a practical way to explore a large number of possible market scenarios and evaluate the resulting distribution of losses.

In this project, a Monte Carlo simulation framework is developed to model portfolio risk using real market data. The focus is not on predicting future prices, but on quantifying potential losses under uncertainty and analyzing how changes in correlation structure affect tail risk.

SECTION 2: SYSTEM DESCRIPTION

In this project, the system under consideration is a portfolio composed of fifteen large-cap Indian equities drawn from different sectors of the economy. The selected companies include **Reliance Industries, Tata Consultancy Services (TCS), Infosys, HDFC Bank, ICICI Bank, Hindustan Unilever, Kotak Mahindra Bank, Larsen & Toubro (L&T), State Bank of India, Bharti Airtel, ITC, Axis Bank, HCL Technologies, Nestle India, and UltraTech Cement**. These firms represent major segments such as energy, information technology, banking and financial services, consumer goods, telecommunications, and infrastructure.

The inclusion of companies from diverse sectors allows the portfolio to exhibit realistic cross-sector interactions and correlation effects. Since these firms are among the most actively traded and capitalized stocks in the Indian market, their price data is reliable and suitable for statistical modeling. The objective is not to analyze firm-specific fundamentals, but to model how their return dynamics interact at the portfolio level.

Historical daily adjusted closing prices are used as the primary data source. Adjusted closing prices are preferred because they account for corporate actions such as dividends and stock splits, ensuring that computed returns reflect true economic gains and losses rather than artificial price changes.

All price series are aligned by date to ensure a common time index across the fifteen assets. Only dates for which data is available for all companies are retained. This alignment step is necessary for accurate estimation of the covariance matrix, which captures the interdependence between asset returns.

For simplicity, the portfolio is constructed using equal weights across all fifteen stocks. This assumption allows the analysis to focus purely on the effect of volatility and correlation on overall portfolio risk, without introducing additional complexity from optimization or dynamic allocation strategies. The portfolio return at each time step is therefore computed as the average of the individual asset returns.

The risk horizon considered in this study is one trading day. Thus, the resulting loss distribution represents possible changes in the total portfolio value over a single day. This short-term horizon is commonly used in risk management and provides a clear and interpretable measure of downside exposure.

SECTION 3: MATHEMATICAL MODELING OF ASSET RETURNS

Although market data is observed in the form of asset prices, prices themselves are not directly suitable for statistical modeling. Asset prices are typically non-stationary and exhibit long-term growth trends, which makes direct probabilistic analysis difficult. For risk analysis, it is more meaningful to study relative price changes rather than absolute price levels.

In this project, asset returns are defined using logarithmic returns. For a given asset with price P_t at time t , the daily log return is defined as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

This formulation measures the proportional change in price from one trading day to the next. Logarithmic returns are preferred because they are additive over time, approximately symmetric for small changes, and independent of the absolute price scale. These properties simplify both statistical estimation and simulation.

Since the portfolio consists of fifteen assets, returns are represented in vector form. At each time step, the return vector is written as:

$$\mathbf{r}_t = \begin{bmatrix} r_{1,t} \\ r_{2,t} \\ r_{3,t} \\ \vdots \\ r_{15,t} \end{bmatrix}$$

where each component corresponds to the daily return of one of the selected companies, such as Reliance, TCS, Infosys, HDFC Bank, and the remaining stocks in the portfolio.

The objective is not to predict future returns but to characterize their statistical behavior. For this purpose, the mean return vector and covariance matrix are estimated from historical return data. The mean vector represents the average daily return of each asset, while the covariance matrix captures both individual asset variability and cross-asset dependence.

The covariance matrix is defined as:

$$\Sigma = \mathbb{E}[(\mathbf{r}_t - \boldsymbol{\mu})(\mathbf{r}_t - \boldsymbol{\mu})^T]$$

where $\boldsymbol{\mu}$ is the mean return vector. For a portfolio of fifteen assets, the covariance matrix is a 15×15 symmetric matrix of the form

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} & \dots & \sigma_{1,15} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} & \dots & \sigma_{2,15} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 & \dots & \sigma_{3,15} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{15,1} & \sigma_{15,2} & \sigma_{15,3} & \dots & \sigma_{15}^2 \end{bmatrix}$$

The diagonal elements σ_i^2 represent the variance of individual asset returns (for example, the variance of Reliance or TCS), while the off-diagonal elements σ_{ij} represent the covariance between pairs of assets. These off-diagonal terms quantify how assets move together. For instance, banking stocks such as HDFC Bank and ICICI Bank may exhibit higher covariance due to sectoral similarities, whereas stocks from unrelated sectors may show lower dependence.

The covariance matrix therefore encodes both individual volatility and inter-asset coupling. This matrix plays a central role in portfolio risk analysis because it determines how uncertainty propagates through the portfolio when assets move simultaneously.

For simulation purposes, asset returns are assumed to follow a multivariate normal distribution characterized by the estimated mean vector μ and covariance matrix Σ . Although real market returns may exhibit deviations such as skewness or heavy tails, this assumption provides a tractable and reasonable first-order approximation for modeling correlated return behavior. Under this formulation, each realization of the return vector represents one possible market outcome for the entire portfolio over a single trading day.

SECTION 4: MONTE CARLO SIMULATION FRAMEWORK

After defining the multivariate return model, the next step is to study how uncertainty in asset returns translates into uncertainty in portfolio losses. Since the portfolio consists of multiple correlated assets, analytical evaluation of the loss distribution becomes difficult, particularly when focusing on extreme outcomes. For this reason, Monte Carlo Simulation is used to approximate the distribution of possible portfolio losses. **The fundamental idea behind Monte Carlo Simulation is to generate a large number of random realizations of asset returns that are consistent with the estimated statistical model.** Each realization represents one possible market scenario for a single trading day. By repeating this process many times, it becomes possible to approximate the full probability distribution of portfolio losses.

To generate correlated asset returns, the simulation begins with independent standard normal random vectors. Let

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_{15} \end{bmatrix}$$

where each $z_i \sim \mathcal{N}(0, 1)$ and the components are independent. These independent random variables are then transformed so that the resulting return vector has the desired covariance structure.

The transformation is performed using the Cholesky decomposition of the covariance matrix. If the covariance matrix Σ is positive definite, it can be written as

$$\Sigma = \mathbf{L}\mathbf{L}^T$$

where $\mathbf{L}$ is a lower triangular matrix. The simulated return vector is then constructed as

$$\mathbf{r}^{(k)} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}^{(k)}$$

where $\boldsymbol{\mu}$ is the mean return vector and $\mathbf{z}^{(k)}$ is the k -th independent random draw. This transformation ensures that the simulated returns have mean $\boldsymbol{\mu}$ and covariance Σ .

For each simulated return vector, the portfolio return is computed as a weighted sum of the individual asset returns. Since equal weights are used in this study, the portfolio return for scenario k is given by

$$R_p^{(k)} = \frac{1}{15} \sum_{i=1}^{15} r_i^{(k)}$$

The corresponding portfolio loss is defined as

$$L^{(k)} = -R_p^{(k)}$$

Defining loss as the negative of return ensures that positive values represent financial losses.

By repeating this procedure for a large number of simulations (e.g., 100,000 scenarios), an empirical distribution of portfolio losses is obtained. This distribution captures not only typical daily fluctuations but also rare and extreme loss events. The resulting loss distribution serves as the foundation for estimating quantitative risk measures such as Value-at-Risk and Expected Shortfall in the subsequent section.

SECTION 5: RISK MEASURES

The Monte Carlo simulation produces a large number of possible portfolio loss realizations. While this empirical loss distribution contains complete information about portfolio risk, it is often necessary to summarize this information using quantitative risk measures that focus specifically on extreme losses. In this project, Value-at-Risk (VaR) and Expected Shortfall (ES) are used for this purpose.

Value-at-Risk is a threshold-based risk measure that identifies a loss level corresponding to a specified confidence level. For a confidence level α , the VaR is defined as the α -quantile of the portfolio loss distribution. Mathematically,

$$VaR_{\alpha} = \inf\{l : P(L \leq l) \geq \alpha\}$$

In practical terms, a 99% VaR represents the loss level that is exceeded only 1% of the time under the assumed model. Thus, VaR provides an estimate of the maximum expected loss over a single day at a given confidence level.

Although VaR indicates where the extreme loss region begins, it does not describe how severe losses can become beyond that threshold. Two portfolios may have identical VaR values while experiencing significantly different magnitudes of loss in extreme scenarios. Therefore, VaR alone does not fully capture tail risk.

To address this limitation, Expected Shortfall is introduced as a complementary risk measure. Expected Shortfall, also referred to as Conditional Value-at-Risk, is defined as the conditional expectation of portfolio loss given that the loss exceeds the VaR level. Mathematically,

$$ES_{\alpha} = \mathbb{E}[L \mid L > VaR_{\alpha}]$$

By construction, Expected Shortfall is always greater than or equal to the corresponding VaR value. While VaR identifies the boundary of extreme losses, Expected Shortfall captures the magnitude of losses within the tail region. For this reason, ES is generally considered a more informative measure of downside risk, especially under stressed conditions.

In this study, both VaR and Expected Shortfall ES are estimated directly from the simulated portfolio loss distribution at the 95% and 99% confidence levels. These measures provide concise numerical summaries of tail risk and form the basis for comparing normal and stressed market scenarios in subsequent sections.

SECTION 6: STRESS TESTING AND CORRELATION EFFECTS

The baseline risk estimates are derived from historical return statistics. However, correlations between assets are not constant and often increase during periods of market stress. When assets begin to move more closely together, the effectiveness of diversification decreases, potentially amplifying portfolio losses.

To study this effect, a correlation stress experiment is conducted by increasing the off-diagonal elements of the covariance matrix while keeping individual asset variances unchanged. This modification isolates the impact of stronger inter-asset dependence without altering standalone volatility levels. The Monte Carlo Simulation is then repeated using the stressed covariance matrix.

Under higher correlation, negative returns across the fifteen stocks tend to occur simultaneously rather than independently. As a result, the portfolio loss distribution shifts toward larger losses and exhibits a heavier tail. Both Value-at-Risk and Expected Shortfall increase under stressed conditions, with Expected Shortfall typically showing a stronger response due to its focus on tail severity.

This analysis demonstrates that portfolio risk is highly sensitive to changes in correlation structure. From a systems viewpoint, stronger correlation acts as tighter coupling between assets, allowing shocks to propagate more directly through the portfolio and reducing diversification benefits.

SECTION 7: SIMULATION PROCEDURE

This section outlines the step-by-step procedure followed to implement the Monte Carlo portfolio risk analysis. The objective is to clearly connect the mathematical return model developed earlier to the simulated portfolio loss distribution used for risk estimation.

The process begins with pre-processing historical data. Adjusted closing prices of the fifteen selected companies are first aligned by date to ensure consistency across assets. From these aligned price series, daily logarithmic returns are computed. These returns serve as the empirical basis for estimating the mean return vector and covariance matrix.

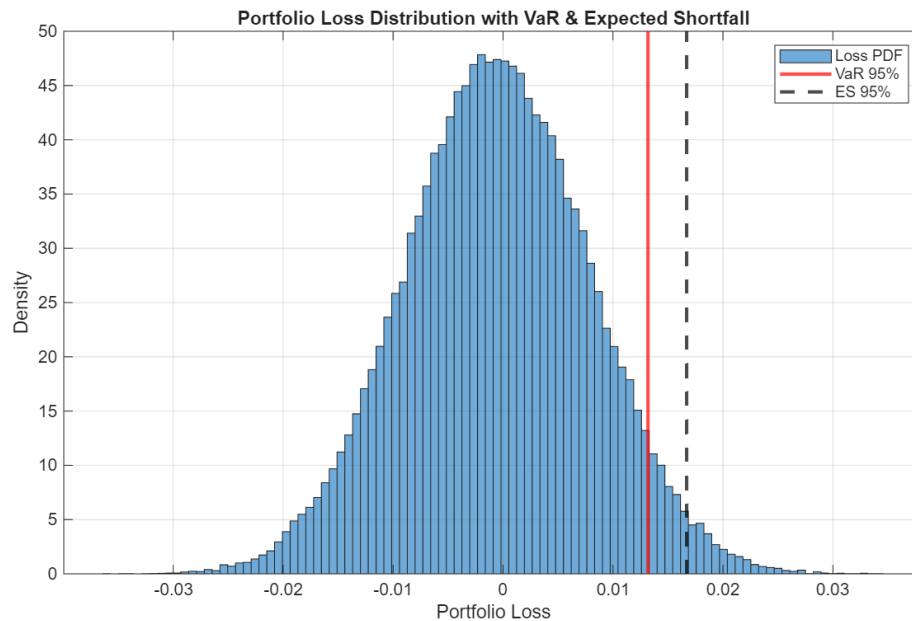
Once the statistical parameters are estimated, the Monte Carlo simulation is performed. Independent standard normal random vectors are generated and transformed using the Cholesky factor of the covariance matrix. This transformation ensures that the simulated asset returns preserve the same dependence structure observed in historical data.

For each simulated return realization, the portfolio return is calculated as the equally weighted average of the fifteen asset returns. The portfolio loss is then defined as the negative of the portfolio return so that positive values correspond to financial losses. Repeating this procedure for a large number of scenarios results in an empirical distribution of one-day portfolio losses.

From this simulated loss distribution, quantitative risk measures such as Value-at-Risk and Expected Shortfall are estimated at the chosen confidence levels. Finally, the simulation is repeated under a modified covariance matrix to evaluate the effect of increased correlation on portfolio risk.

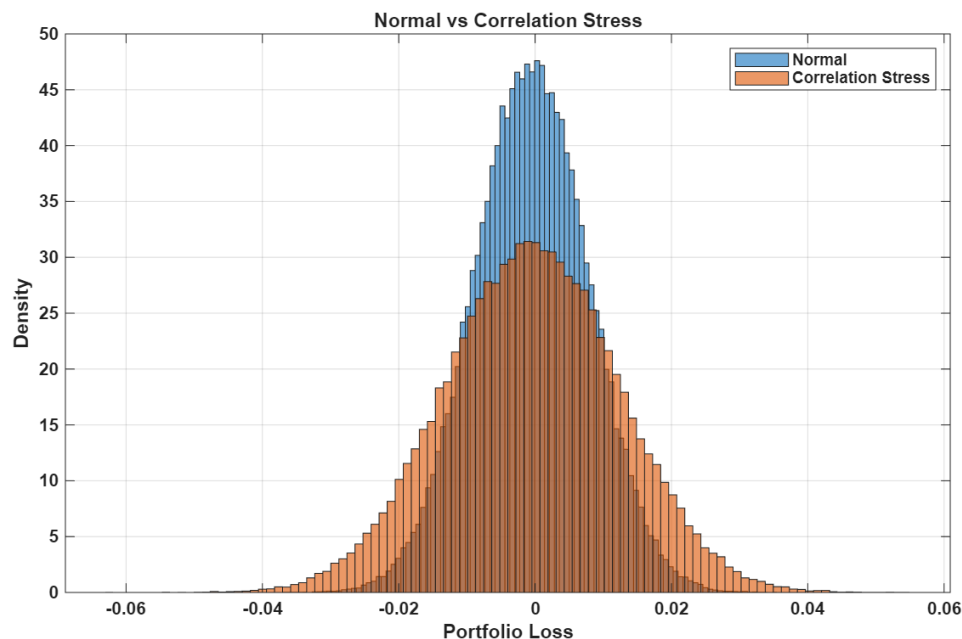
SECTION 8: RESULTS

Figure 1:



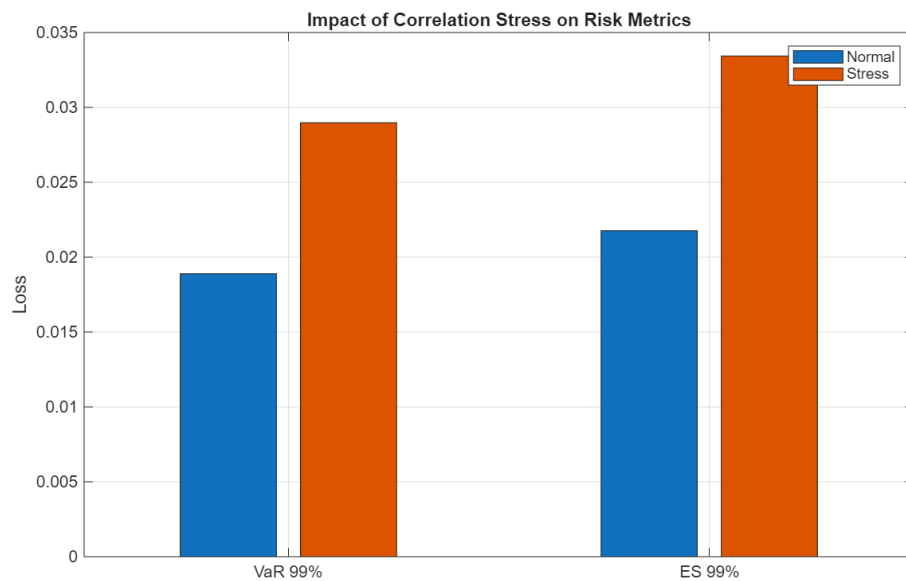
As shown in Figure 1 shows the simulated one-day loss distribution of the fifteen-stock portfolio obtained from Monte Carlo simulation. Most outcomes are clustered around small losses, reflecting typical market fluctuations. The red line indicates the 95% Value-at-Risk, marking the loss level exceeded in only 5% of cases, while the dashed black line represents the 95% Expected Shortfall, capturing the average severity of those extreme losses beyond the VaR threshold. **The fact that Expected Shortfall (ES) lies further in the right tail highlights that losses beyond the Value-at-Risk (VaR) threshold can be significantly larger.**

Figure 2:



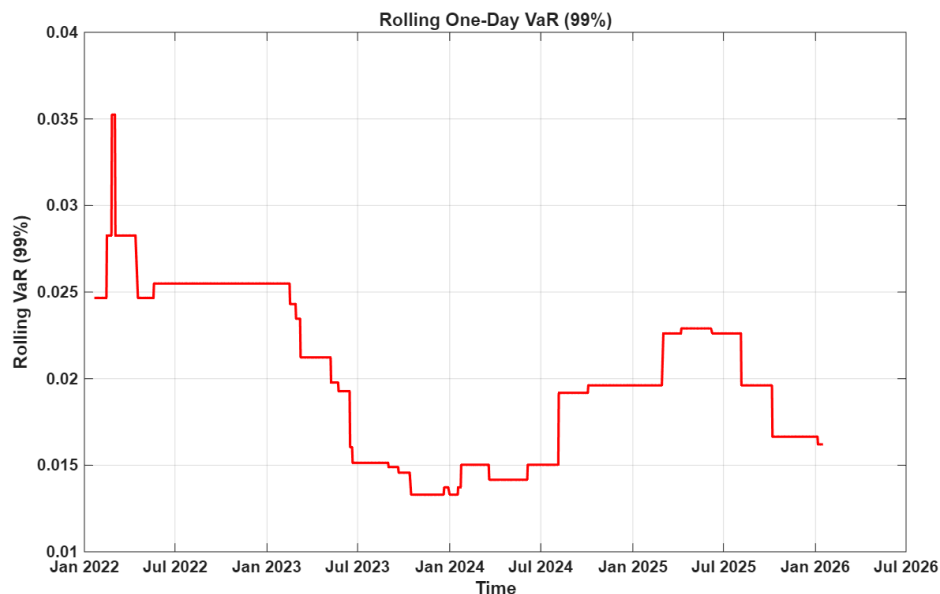
As shown in Figure 2 compares the portfolio loss distributions under normal historical correlation and increased correlation stress. **Under stressed conditions, the distribution becomes wider and shifts toward larger losses, indicating that assets move more closely together. This reduction in diversification leads to a heavier right tail, reflecting a higher probability of extreme portfolio losses.**

Figure 3:



As shown in Figure 3 compares the 99% Value-at-Risk and Expected Shortfall under normal and stressed correlation conditions. Both risk measures increase significantly when correlation is elevated, indicating higher downside exposure. **The larger increase in Expected Shortfall highlights that extreme losses become more severe under stress, emphasizing the sensitivity of tail risk to changes in inter-asset dependence.**

Figure 4:



As shown in Figure 4 the rolling one-day 99% Value-at-Risk of the fifteen-stock Indian equity portfolio computed using a 250-day moving window. The variation in the curve indicates that portfolio risk is time-varying rather than constant. **The elevated VaR observed in early 2022 reflects a period of increased market volatility driven by global macroeconomic uncertainty, inflationary pressures, and geopolitical tensions such as the Russia–Ukraine conflict. During such periods, asset volatility and cross-asset correlation tend to increase, leading to higher estimated downside risk.** Overall, this graph demonstrates that portfolio risk evolves over time and highlights the importance of dynamic risk monitoring rather than relying on static estimates.

SECTION 9: CONCLUSION

In this project, a Monte Carlo–based framework was developed to quantify portfolio risk using real market data from fifteen large-cap Indian equities. Asset returns were modeled as a multivariate stochastic process characterized by an empirically estimated mean vector and covariance matrix. Using simulation, a large number of possible one-day market scenarios were generated to obtain an empirical distribution of portfolio losses.

From this distribution, Value-at-Risk and Expected Shortfall were computed to measure downside risk. The results show that while most daily outcomes remain moderate, extreme losses in the tail of the distribution can be significant. Expected Shortfall consistently exceeds Value-at-Risk, highlighting the importance of analyzing tail severity rather than relying solely on threshold-based measures.

The correlation stress experiment further demonstrated that portfolio risk is highly sensitive to inter-asset dependence. When correlations increase, diversification benefits weaken and tail losses become more severe. The rolling VaR analysis confirmed that risk is time-varying and influenced by changing market conditions.

Overall, this study illustrates how stochastic modeling and Monte Carlo simulation can be used to systematically analyze portfolio risk under both normal and stressed environments. The framework developed in this project provides a practical and interpretable approach to understanding downside exposure in a multi-asset portfolio.

SECTION 10: PRACTICAL RELEVANCE AND APPLICATIONS

The framework developed in this project has direct relevance in practical risk management and financial decision-making. Financial institutions such as banks, asset management firms, and hedge funds routinely use simulation-based methods to estimate potential portfolio losses under uncertainty. The Monte Carlo approach implemented here mirrors the core structure of internal risk engines used for capital allocation and regulatory reporting.

Value-at-Risk and Expected Shortfall are widely used metrics in industry for assessing downside exposure. These measures are used to determine capital reserves, set risk limits, and monitor trading desks. The ability to simulate correlated asset returns and quantify tail losses is essential for understanding systemic risk within diversified portfolios.

The correlation stress experiment performed in this study is particularly relevant during periods of market instability. In practice, risk managers routinely conduct stress testing to evaluate how portfolios behave under extreme but plausible scenarios. The results of such stress tests influence hedging decisions, portfolio rebalancing strategies, and risk budgeting.

The rolling VaR analysis further demonstrates the importance of dynamic risk monitoring. In real-world settings, risk levels are continuously updated based on evolving market conditions. Time-varying risk estimates help institutions respond proactively to volatility spikes rather than relying on static historical measures.

Beyond financial institutions, the concepts applied in this project — stochastic modeling, covariance analysis, and Monte Carlo simulation — are broadly applicable in engineering, insurance, and operations research, where uncertainty quantification plays a central role in system design and decision-making.

SECTION 11: MATLAB IMPLEMENTATION

The complete MATLAB script used to implement the Monte Carlo portfolio risk engine is provided below. The code performs data pre-processing, return computation, covariance estimation, correlated scenario generation, portfolio loss calculation, risk metric estimation, and stress testing.

```
1 % MONTE CARLO PORTFOLIO RISK ENGINE
2 clc; clear; close all;
3 % FILE LIST
4 files = {
5     'RELIANCE.NS.csv', 'TCS.NS.csv', 'INFY.NS.csv', 'HDFCBANK.NS.csv', ...
6     'ICICIBANK.NS.csv', 'HINDUNILVR.NS.csv', 'KOTAKBANK.NS.csv', 'LT.NS.csv', ...
7     'SBIN.NS.csv', 'BHARTIARTL.NS.csv', 'ITC.NS.csv', 'AXISBANK.NS.csv', ...
8     'HCLTECH.NS.csv', 'NESTLEIND.NS.csv', 'ULTRACEMCO.NS.csv'
9 };
10 numAssets = length(files);
11 % LOAD & ALIGN DATA
12 priceTable = table();
13 for i = 1:numAssets
14
15     if ~isfile(files{i})
16         error('File NOT found: %s', files{i});
17     end
18     T = readtable(files{i}, 'VariableNamingRule','preserve');
19     T = T(:, {'Date','Adj Close'});
20     stockName = erase(files{i}, '.NS.csv');
21     T.Properties.VariableNames{'Adj Close'} = stockName;
22
23     if i == 1
24         priceTable = T;
25     else
26         priceTable = innerjoin(priceTable, T, 'Keys','Date');
27     end
28 end
29 disp('All stock prices aligned successfully.');
```

```
30 % RETURN MODELING
31 priceMat = priceTable(:,2:end);
32 logPrices = log(priceMat);
33 returns = diff(logPrices);
34 mu = mean(returns)';
35 Sigma = cov(returns);
36 % MONTE CARLO ENGINE
37 numSim = 100000;
38 L = chol(Sigma,'lower');
39 Z = randn(numAssets, numSim);
40 simReturns = mu + L * Z;
41 weights = ones(numAssets,1) / numAssets;
42 portfolioLosses = -(weights' * simReturns);
43 % VaR & EXPECTED SHORTFALL
44 VaR_95 = quantile(portfolioLosses, 0.95);
45 VaR_99 = quantile(portfolioLosses, 0.99);
46
47 ES_95 = mean(portfolioLosses(portfolioLosses > VaR_95));
48 ES_99 = mean(portfolioLosses(portfolioLosses > VaR_99));
49
50 fprintf('\n--- PORTFOLIO RISK METRICS ---\n');
51 fprintf('95%% VaR = %.4f | 95%% ES = %.4f\n', VaR_95, ES_95);
52 fprintf('99%% VaR = %.4f | 99%% ES = %.4f\n', VaR_99, ES_99);
53
54 % LOSS DISTRIBUTION
55 figure;
56 histogram(portfolioLosses,100,'Normalization','pdf');
57 hold on;
58 xline(VaR_95,'r','LineWidth',2);
59 xline(ES_95,'k--','LineWidth',2);
60 xlabel('Portfolio Loss');
61 ylabel('Density');
```

```

62 title('Portfolio Loss Distribution with VaR & Expected Shortfall');
63 legend('Loss PDF', 'VaR 95%', 'ES 95%');
64 grid on;
65 % CORRELATION STRESS TEST
66 rho = 0.8;
67 Sigma_stress = Sigma;
68
69 for i = 1:numAssets
70     for j = 1:numAssets
71         if i ~= j
72             Sigma_stress(i,j) = rho * sqrt(Sigma(i,i)*Sigma(j,j));
73         end
74     end
75 end
76 L_stress = chol(Sigma_stress, 'lower');
77 loss_stress = -(weights' * (mu + L_stress * Z));
78
79 % NORMAL vs STRESS
80 figure;
81 histogram(portfolioLosses, 100, 'Normalization', 'pdf');
82 hold on;
83 histogram(loss_stress, 100, 'Normalization', 'pdf');
84 xlabel('Portfolio Loss');
85 ylabel('Density');
86 title('Normal vs Correlation Stress');
87 legend('Normal', 'Correlation Stress');
88 grid on;
89
90 % VaR & ES COMPARISON
91 VaR_99_stress = quantile(loss_stress, 0.99);
92 ES_99_stress = mean(loss_stress(loss_stress > VaR_99_stress));
93
94 figure;
95 bar([VaR_99, VaR_99_stress; ES_99, ES_99_stress]);
96 set(gca, 'XTickLabel', {'VaR 99%', 'ES 99%'});
97 ylabel('Loss');
98 title('Impact of Correlation Stress on Risk Metrics');
99 legend('Normal', 'Stress');
100 grid on;
101
102 % ROLLING VaR
103 window = 250;
104 portfolioReturnsHist = -(returns * weights);
105
106 rollingVaR = zeros(length(portfolioReturnsHist)-window, 1);
107
108 for t = window+1:length(portfolioReturnsHist)
109     rollingVaR(t-window) = quantile(portfolioReturnsHist(t-window:t-1), 0.99);
110 end
111
112 datesRolling = priceTable.Date(window+2:end);
113 figure;
114 plot(datesRolling, rollingVaR, 'LineWidth', 1.5);
115 xlabel('Time');
116 ylabel('Rolling VaR (99%)');
117 title('Rolling One-Day VaR (99%)');
118 grid on;

```

REFERENCES:

- [1] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th ed. Pearson, 2018.
- [2] P. Jorion, *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed. McGraw-Hill, 2007.
- [3] A. J. McNeil, R. Frey, and P. Embrechts, *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton, NJ, USA: Princeton University Press, 2015.
- [4] P. Glasserman, *Monte Carlo Methods in Financial Engineering*. New York, NY, USA: Springer, 2004.
- [5] S. M. Ross, *Simulation*, 5th ed. Amsterdam, The Netherlands: Academic Press, 2012.
- [6] H. Markowitz, “Portfolio selection,” *J. Finance*, vol. 7, no. 1, pp. 77–91, Mar. 1952.
- [7] R. T. Rockafellar and S. Uryasev, “Optimization of conditional value-at-risk,” *J. Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [8] P. Artzner, F. Delbaen, J.-M. Eber, and D. Heath, “Coherent measures of risk,” *Math. Finance*, vol. 9, no. 3, pp. 203–228, Jul. 1999.
- [9] R. Cont, “Empirical properties of asset returns: Stylized facts and statistical issues,” *Quant. Finance*, vol. 1, no. 2, pp. 223–236, 2001.
- [10] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *J. Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [11] F. Black and M. Scholes, “The pricing of options and corporate liabilities,” *J. Polit. Econ.*, vol. 81, no. 3, pp. 637–654, 1973.
- [12] D. Duffie and K. J. Singleton, *Credit Risk: Pricing, Measurement, and Management*. Princeton, NJ, USA: Princeton University Press, 2003.
- [13] D. Ruppert, *Statistics and Data Analysis for Financial Engineering*, 2nd ed. New York, NY, USA: Springer, 2015.
- [14] Bank for International Settlements, “Basel III: International framework for liquidity risk measurement, standards and monitoring,” Basel, Switzerland, 2010.
- [15] B. B. Mandelbrot, “The variation of certain speculative prices,” *J. Business*, vol. 36, no. 4, pp. 394–419, Oct. 1963.